

Sri Sivasubramaniya Nadar College of Engineering, Chennai

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
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Experiment 8: Effect of PCA on Regression and Classification Models

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1 Objective

- To study the impact of Principal Component Analysis (PCA) on machine learning models.
- To compare performance with and without PCA.
- To evaluate models using 5-fold cross-validation.

2 Dataset

Dataset: PCA Full Lab Dataset (1000 samples)

- Features: Academic and performance attributes
- Regression Target: Final Score
- Classification Target: Performance Level

3 Theory

PCA is a dimensionality reduction technique that transforms features into orthogonal components maximizing variance.

- Reduces multicollinearity
- Improves generalization
- Removes noise

4 Exploratory Data Analysis

4.1 Feature Distribution

Figure 1: Feature Distribution Histograms

4.2 Class Distribution

Figure 2: Class Distribution

4.3 Comparisons

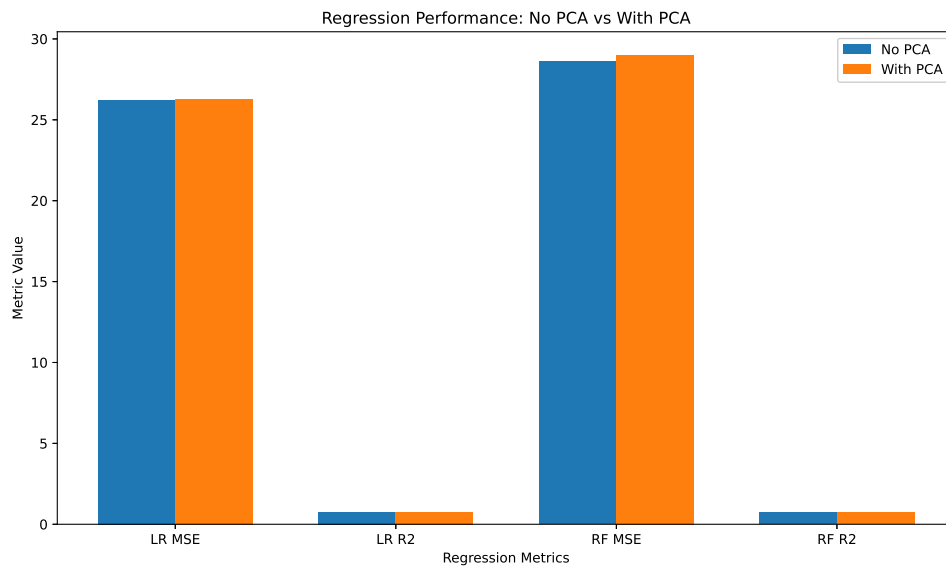


Figure 3: Regression Comparison

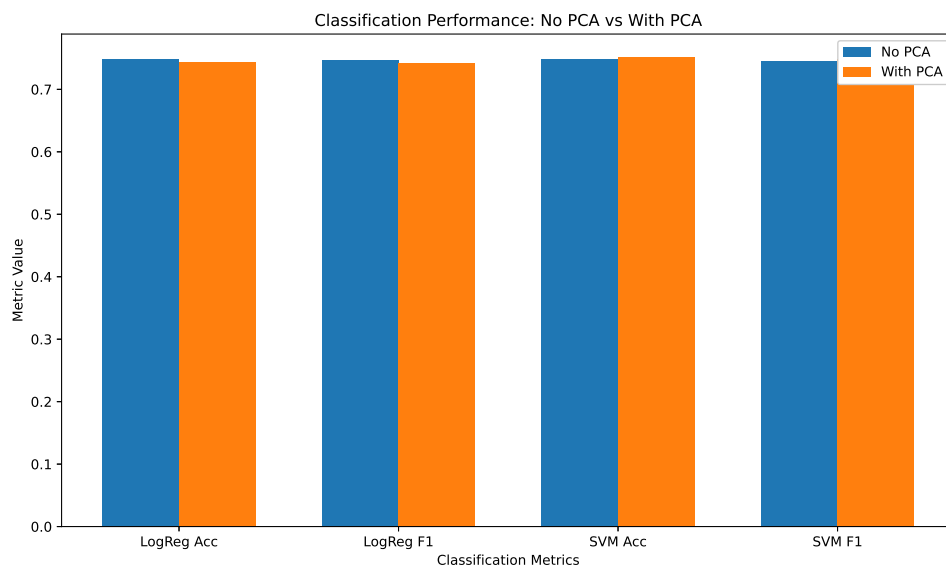


Figure 4: Classification Comparison

5 Preprocessing Steps

- Handled missing values
- Encoded categorical target
- Standardized features
- Applied 5-fold cross-validation

6 PCA Implementation

- Applied PCA after standardization
- Retained 95% variance
- Generated scree plot

Table 1: PCA Summary

Components	Explained Variance (%)	Justification
7	95.8952	retains 95% cumulative variance

7 Regression Models

7.1 Models Used

- Linear Regression
- Random Forest Regressor

Table 2: Regression Results (5-Fold CV)

Model	Metric	No PCA	With PCA	Observation
Linear Regression	MSE	26.2338402	26.2766122	marginal increase
Linear Regression	R2	0.7479	0.7476	marginal decrease
Random Forest	MSE	28.6163	28.9935	Increase
Random Forest	R2	0.7254	0.7217	increase

8 Classification Models

8.1 Models Used

- Logistic Regression
- Support Vector Machine (SVM)

Table 3: Classification Results (5-Fold CV)

Model	Metric	No PCA	With PCA	Observation
Logistic Regression	Accuracy	0.749	0.744	marginal decrease
Logistic Regression	F1-score	0.746	0.741	decrease
SVM	Accuracy	0.748	0.751	increase
SVM	F1-score	0.7445	0.748	increase

9 Analysis Questions

1. PCA improves Linear Regression as it reduces multicollinearity.
2. Random Forest is less affected as it is tree-based.
3. Logistic Regression benefits more from PCA.
4. PCA reduces variance across folds.
5. Linear models benefit more due to feature decorrelation.

10 Conclusion

- PCA is useful for linear models and high-dimensional data.
- Not necessary for tree-based models.
- Trade-off exists between interpretability and performance.